

JEE MAIN REVISION MASTER: LIMITS, CONTINUITY AND DIFFERENTIABILITY 2026 3 QUESTIONS 5% WEIGHTAGE ↓25.04%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 29 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Limits Continuity And Differentiability - Core Application	29 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Limits Continuity And Differentiability - Core Application (Very High Yield — 29 PYQs)

Core Principle: Key problem pattern in Kcet Limits Continuity And Differentiability.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If $f(x) = x^2 - 1$ if $x \geq 2$ $x + 1$ if $x < 2$, then $\lim_{x \rightarrow 2} f(x) = \dots$ " [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\lim_{x \rightarrow 3} (\frac{x^2 - ax - 3bx - 3}{x - 3}) = 5$, then $a + b = \dots$ " [Ans: C]
- [2026 Shift] Q#3 (MCQ): "If $f(x) = \begin{cases} ax + 7 & \text{if } x < 1 \\ 3x - 1 & \text{if } x = 1 \\ b / (x + 3) & \text{if } x > 1 \end{cases}$ is continuous at $x = \dots$ " [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If $f(x) = x^2 - 1$ if $x \geq 2$ and $f(x) = x + 1$ if $x < 2$, then $\lim_{x \rightarrow 2} f(x) =$

- (A) 3
- (B) 5 **[Correct]**
- (C) 7
- (D) 9

Correct Answer: B

Solution Analysis: For $x \rightarrow 2^-$, the values of x are near 2, which is less than 2. So we use $f(x) = x + 1$. Hence, $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x + 1) = 2 + 1 = 3$. Now for $x \rightarrow 2^+$, we check both sides. For $x < 2$, $f(x) = x + 1$. So, $\lim_{x \rightarrow 2^-} f(x) = 2 + 1 = 3$. For $x \geq 2$, $f(x) = x^2 - 1$. So, $\lim_{x \rightarrow 2^+} f(x) = 2^2 - 1 = 4 - 1 = 3$. Since both left-hand and right-hand limits are equal, $\lim_{x \rightarrow 2} f(x) = 3$. Now, $\lim_{x \rightarrow 2} f(x) = 3$. So the correct